Couette flow

Wiki *Enhanced dissipation for passive scalars*

8 November 2026

Page team: to be named · Status: skeleton
Discuss on Zulip: #examples, topic "Couette flow".

[Lead: what this example shows, in three lines.]

1 Setting

[The flow $u = (y, 0)$ on $\mathbb{T} \times \mathbb{R}$: the equation, the Fourier modes, what decays.]

2 The result

Theorem 1 (enhanced dissipation for Couette flow)**.** *[The statement.]*

Lean: not yet formalized.

Idea. [The key mechanism in a few lines.]

Proof architecture. [The proof as a short chain of steps, and its diagram.]

3 One example, every technique

Each proof below is written for this example only, to show the mechanism; the general results, with complete proofs, are in the part *Techniques and general results*.

3.1 Explicit solution

Proof by the explicit solution. [The proof.] □

3.2 Spectral methods

Proof by spectral methods. [The proof.] □

General theory: Spectral methods.

3.3 Hypocoercivity

Proof by hypocoercivity. [The proof.] □

General theory: Hypocoercivity.

3.4 Stochastic methods

Proof by stochastic methods. [The proof.] □

General theory: Stochastic methods.

3.5 Hörmander theory

Proof by Hörmander theory. [The proof.] □

General theory: Hörmander theory.

4 What this example teaches

[What the example shows, and what the general theory keeps of it.]

References